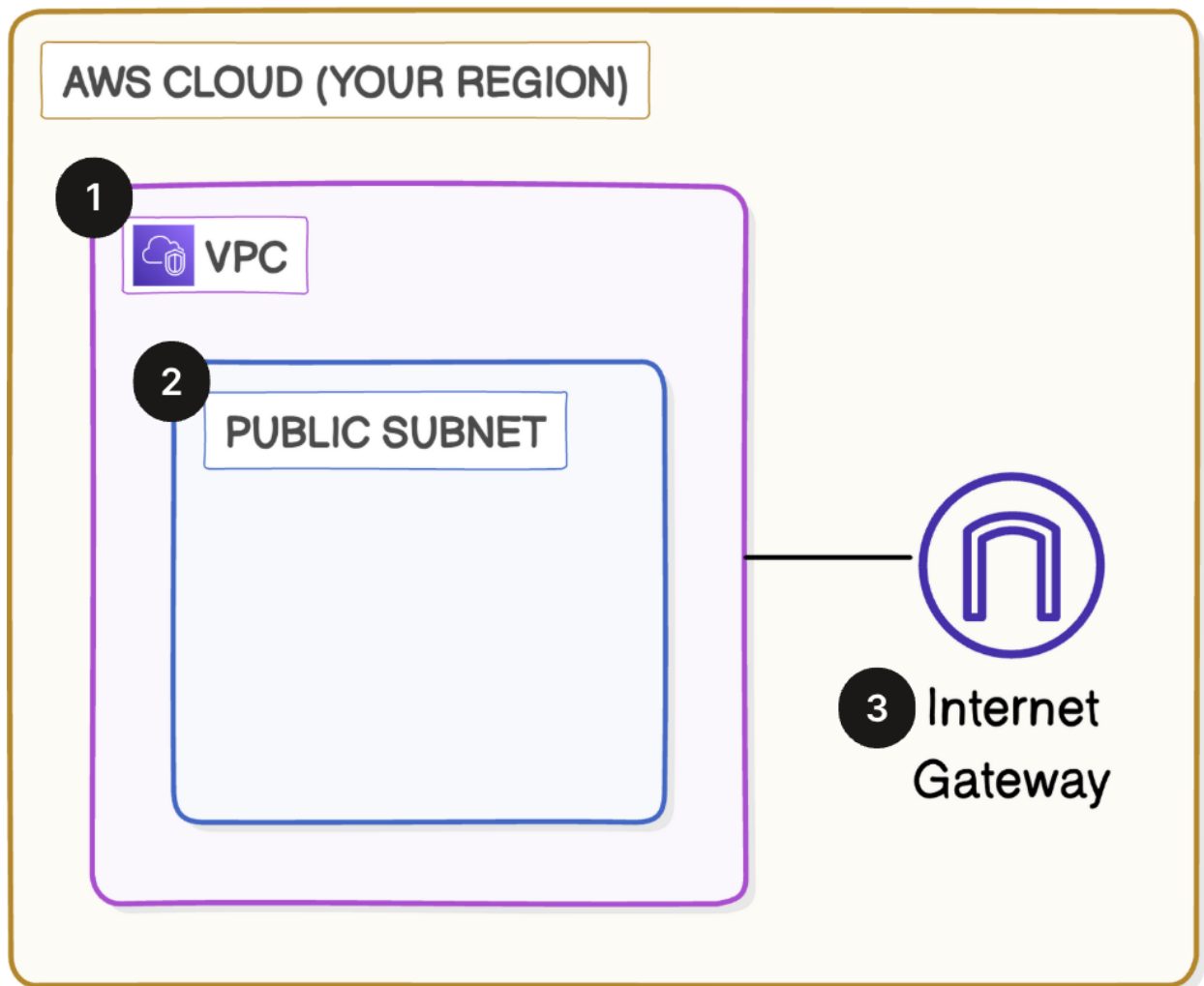


First of all when we think of VPC what does it come first like it is like sort of virtual cloud, well it's called Virtual Private Cloud, it's like a private city in country. You can design your own connection like subnets, traffic rules, or security measures

We will

- Create Amazon VPC
- Create Public Subnet
- Create Internet Gateway ‘



- Subnets are basically segment your network

Now we will create VPC

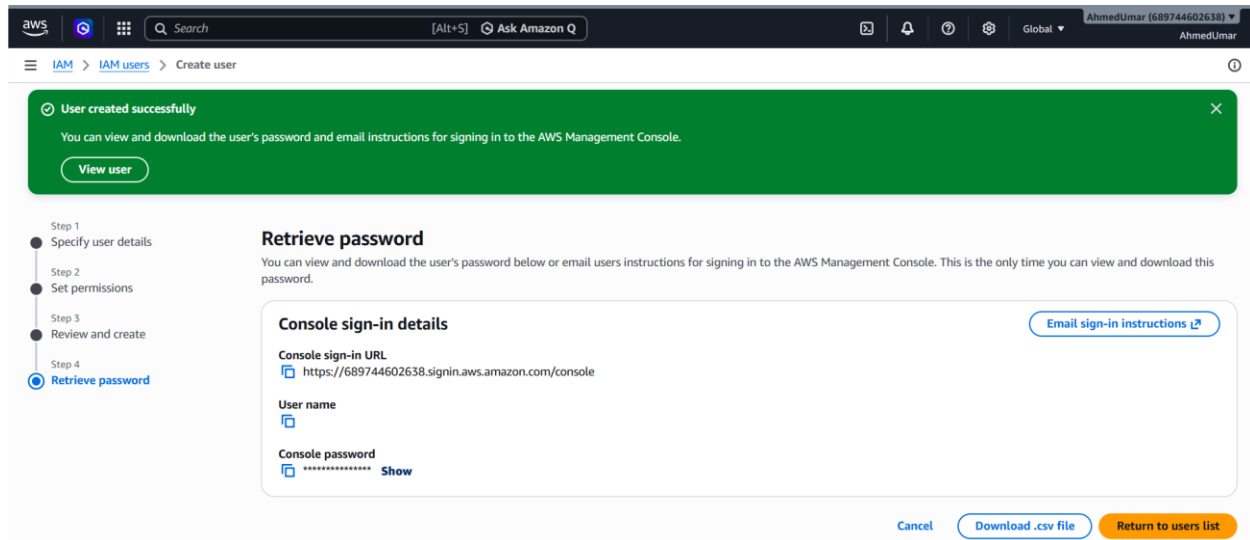
Now for this, you have to make IAM Users not Root account now for this we have to make that first

The screenshot shows the AWS IAM console interface. The top navigation bar includes the AWS logo, a search bar, and the user's name 'AhmedUmar (689744602638)'. The breadcrumb trail is 'IAM > IAM users > Create user'. On the left, a sidebar shows the steps: 'Step 1: Specify user details' (selected), 'Step 2: Set permissions', and 'Step 3: Review and create'. The main content area is titled 'Specify user details' and contains a 'User details' section. The 'User name' field is filled with 'ahmedumar-iam-admin'. Below it, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, =, @, _ - (hyphen)'. There is an unchecked checkbox for 'Provide user access to the AWS Management Console - optional' with a note: 'In addition to console access, users with SigninLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.' At the bottom right of the form are 'Cancel' and 'Next' buttons.

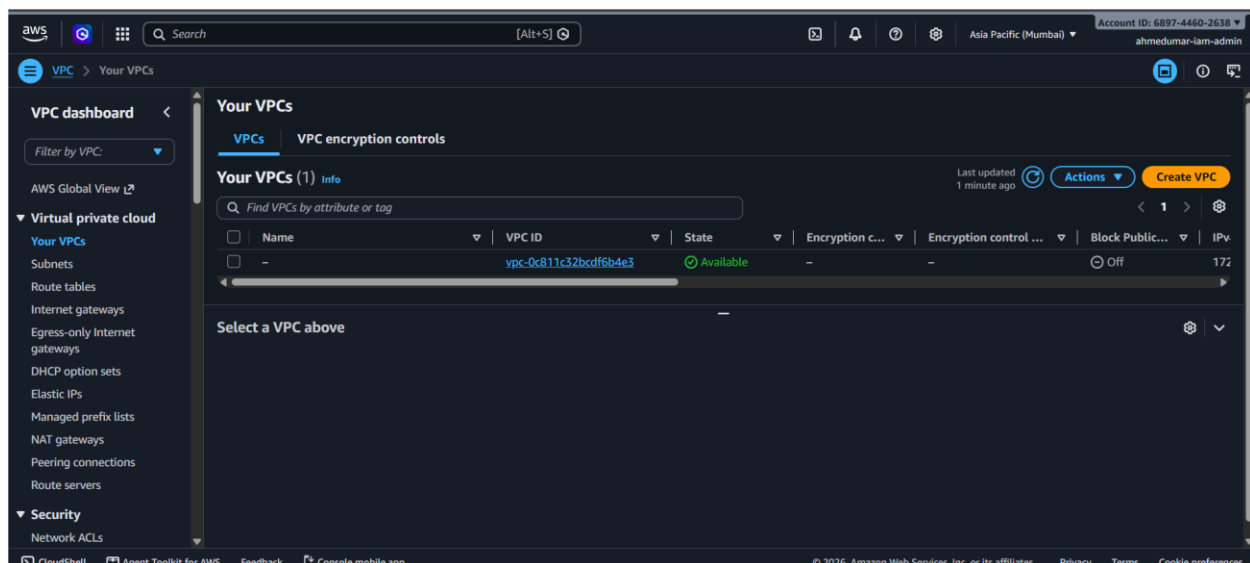
- Go to IAM-> Users, and then click create user and fill details according to your needs
- Also select the check box “Provider User Access.....”

The screenshot shows the 'Console password' section of the 'Create user' process. It has two radio button options: 'Autogenerated password' (unselected) and 'Custom password' (selected). Below the 'Custom password' option is a text input field containing eight dots. To the right of the input field, there are two bullet points: 'Must be at least 8 characters long' and 'Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' '. Below the input field is an unchecked checkbox for 'Show password'. At the bottom, there is an unchecked checkbox for 'Users must create a new password at next sign-in - Recommended' with a note: 'Users automatically get the IAMUserChangePassword policy to allow them to change their own password.' At the bottom right of the form is a blue information box with a note: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. Learn more'. At the bottom right of the form are 'Cancel' and 'Next' buttons.

- Deselect that check box also
- Then choose Next, Next and then click Create User



- You have created your user
- Now login using Username and Console Password as IAM user
- NOTE: add this permission AmazonVPCFullAccess during making user



- Now, you will see default VPC
- Now click create VPC

aws [Search] [Alt+S]

VPC > Your VPCs > Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

Network VPC

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

- Choose VPC only
- Name tag
- Enter 10.0.0.0/16 IPv4

💡 What does IPv4 mean?

IPv4 stands for Internet Protocol version 4, which is the most common way to write an IP address. IPv4 address are written as four sets of numbers separated by dots (e.g., 192.168.0.1).

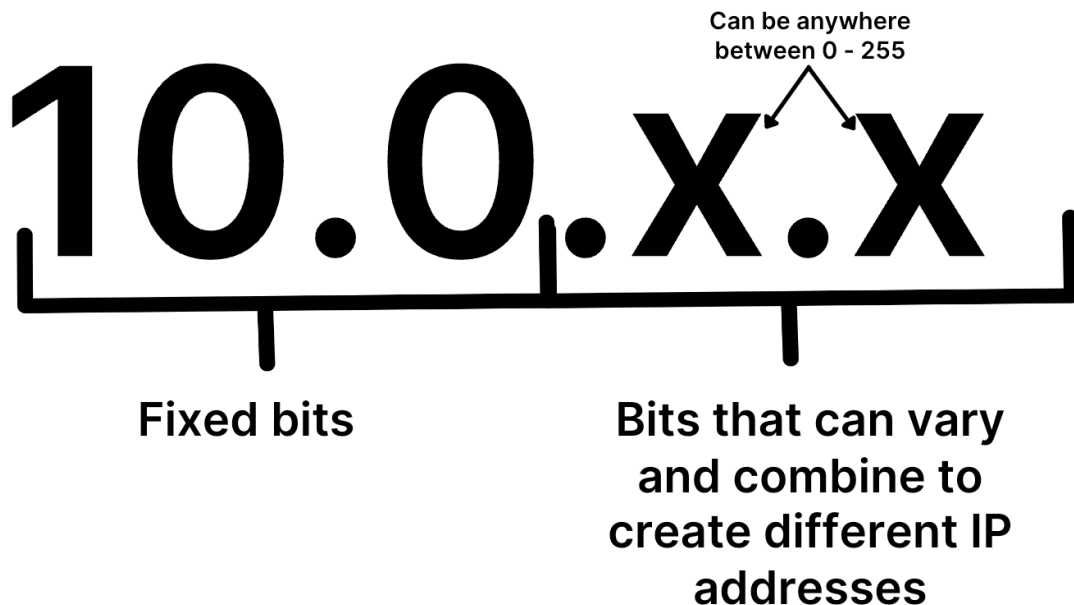
💡 What is a CIDR block?

CIDR (which stands for Classless Inter-Domain Routing) is a way to assign a whole block of IP addresses, kind of like creating a zone/area in a city.

To understand how big a CIDR block is, look at the number after the slash - the **smaller** the number, the larger the CIDR block!

For example, `10.0.0.0/16` means the first 16 bits of your IP address (`10.0`) are fixed, but the remaining 16 bits (i.e. the second half of the IP address) can be allocated however you like. Addresses within this CIDR block start at `10.0.0.0` and go up to `10.0.255.255`. There are 2^{16} (65,536) possible IP addresses within this subnet.

10.0.0.0/16 is a CIDR block with 65,536 possible IP addresses



CIDR block	Number of fixed bits	Number of bits that vary	IP addresses range between
10.0.0.0/16	16	16	10.0.0.0 to 10.0.255.255
10.0.0.0/24	24	8	10.0.0.0 to 10.0.0.255
10.0.0.0/8	8	24	10.0.0.0 to 10.255.255.255

- Now We will create Subnets inside VPC

AWS CLOUD (YOUR REGION)



VPC

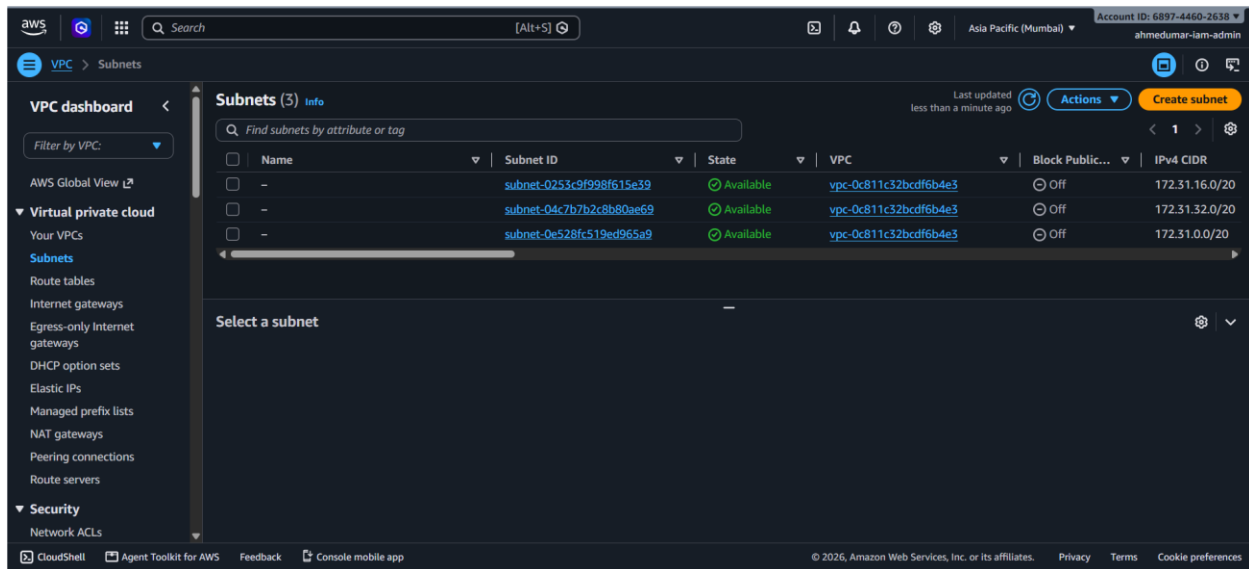
10.0.0.0/16

PUBLIC SUBNET

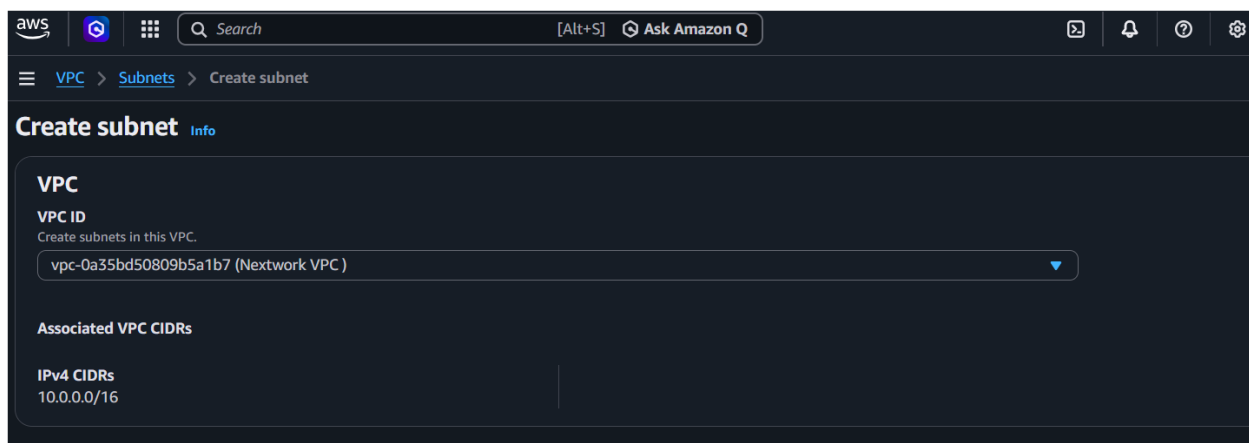
10.0.0.0/24

You
are here!





- Now go to subnets
- You will see a lot of subnets(joking just 3)
- You use subnets to group resources with similar access rules and restrictions.
- Now choose Create Subnet



Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

- Configure important settings and so fill according to screenshots in order not to miss any configurations

You have successfully created 1 subnet: subnet-09cc65f66a4a5ed97

Subnets (1) [Info](#)

Last updated 1 minute ago [Actions](#) [Create subnet](#)

Subnet ID : subnet-09cc65f66a4a5ed97

Clear filters

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	Public 1	subnet-09cc65f66a4a5ed97	Available	vpc-0a35bd50809b5a1b7 Nex...	Off	10.0.0.0/24

Select a subnet

- Your Subnet has been created
- Now select Public 1 and then select edit subnet settings

☰ VPC > Subnets > subnet-09cc65f66a4a5ed97 > Edit subnet settings ⓘ

Edit subnet settings ⓘ

Subnet

Subnet ID ☐ subnet-09cc65f66a4a5ed97	Name ☐ Public 1
--	---------------------------

Auto-assign IP settings ⓘ
Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ **Enable auto-assign public IPv4 address ⓘ**

☐ **Enable auto-assign customer-owned IPv4 address ⓘ**
Option disabled because no customer owned pools found.

Resource-based name (RBN) settings ⓘ
Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ **Enable resource name DNS A record on launch ⓘ**

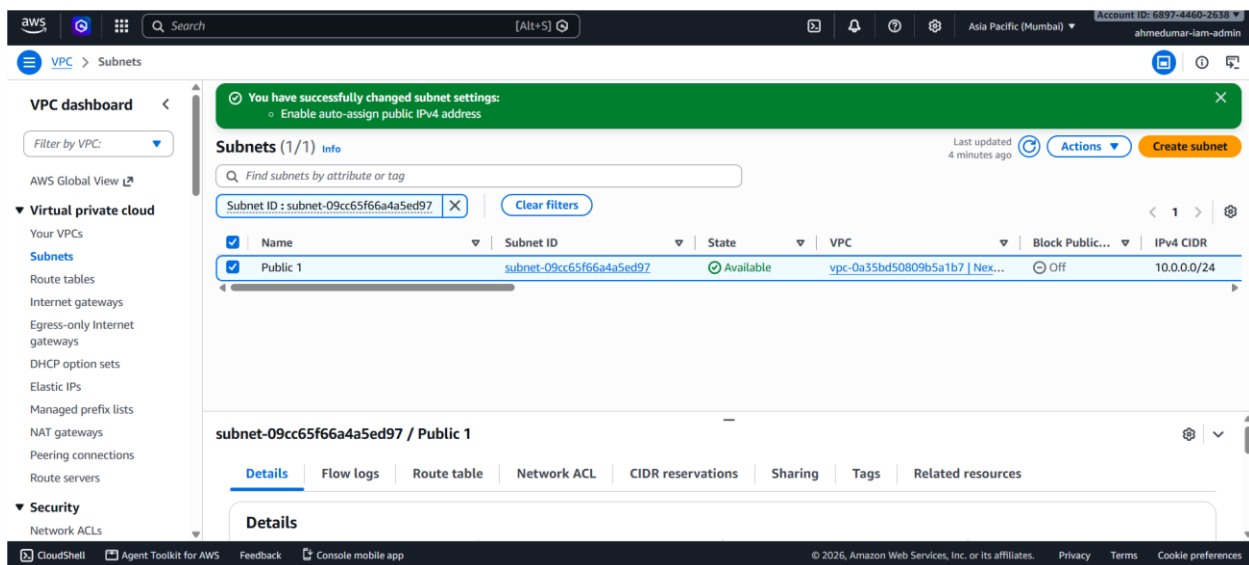
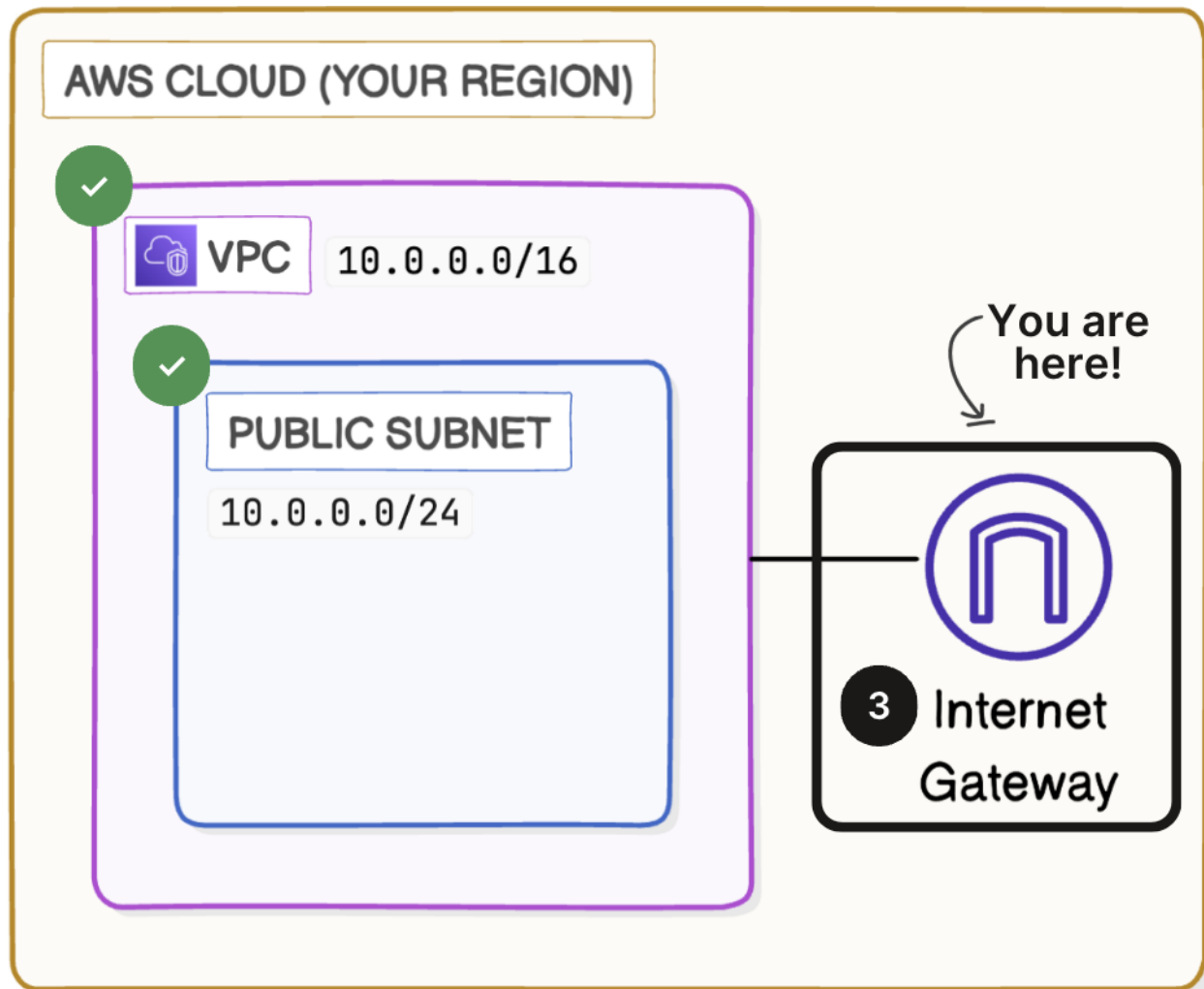
☐ **Enable resource name DNS AAAA record on launch ⓘ**

Hostname type ⓘ

☐ Resource name

☒ IP name

- Tick Enable auto-assign Checkbox
 - Click Save
-
- Now we will create internet gateway
 - Connect your VPC to the internet using a internet gateway.



- Choose internet gateways

- Then click create internet gateways

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

NextWork IG

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	NextWork IG	Remove

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

- Fill all that necessary information
- Click create internet gateway

Attach to VPC (igw-0456523a4804b7dce) [Info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

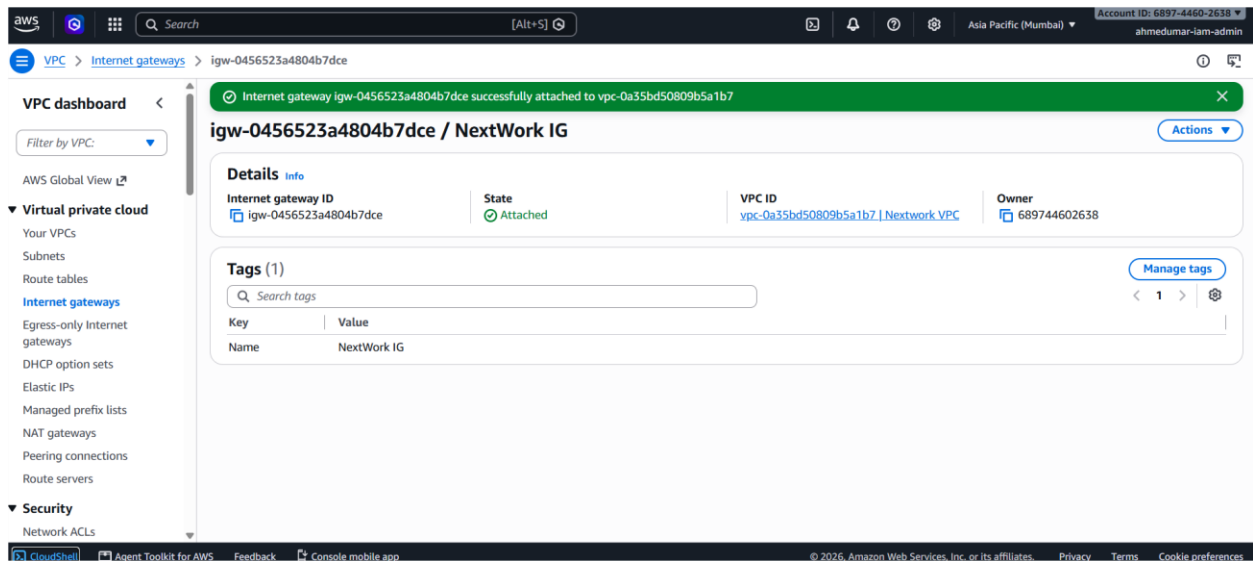
vpc-0a35bd50809b5a1b7

[AWS Command Line Interface command](#)

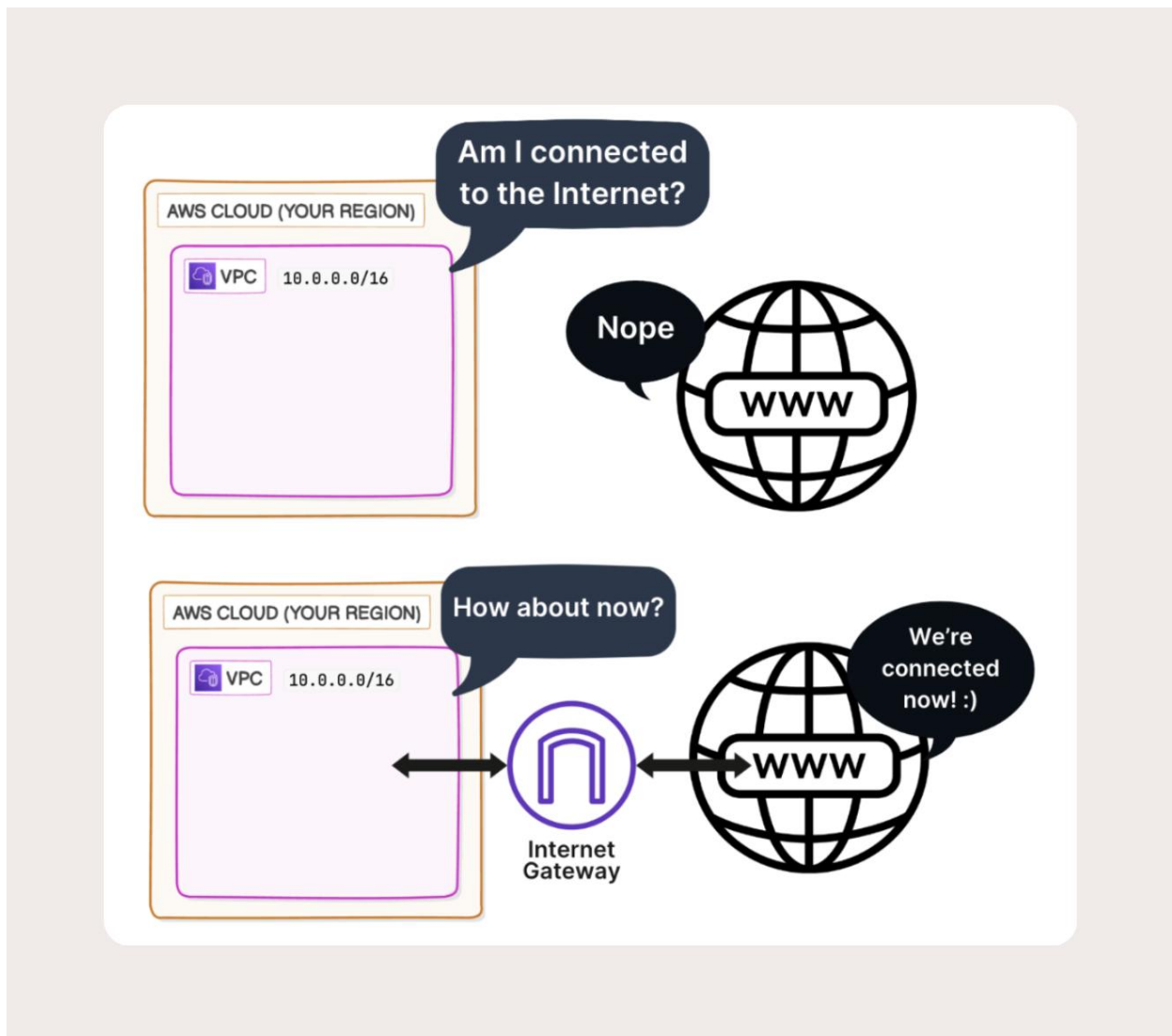
[Cancel](#) [Attach internet gateway](#)

- Select your network
- Then go to actions
- Select attach to VPC
- Select VPC which we created earlier

- Then click *attach internet gateway*



- This message means that:



You made your first networking project

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